York University & LMU Munich

Virtual Journal Club

Grand Proposal

AIfication of society: How AI changes our decision making

An experimental approach

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1 Executive Summary

Artificial intelligence is a constant presence in our lives. From virtual assistants to customer service bots, AI helps guide daily choices. Experts estimate that over half of adults in developed countries use AI every day (McKinsey & Company, 2023). This number is expected to rise further.

Given this, the question of whether AI actually helps us make better choices becomes more pressing. If AI sharpens our thinking, it can support fairer, faster decisions. But if it leads us to trust blindly, it risks doing more harm than good.

Existing economic literature offers two opposing views. Daniel Kahneman's dual-system theory states we think in two modes: fast and intuitive (System 1), or slow and logical (System 2). AI could act as a cognitive aid. This could improve decision quality by encouraging deeper, System 2 thinking. Conversely, Herbert Simon's theory of bounded rationality emphasizes our mental limits. Even with AI, these remain. Worse, AI might overwhelm us with information, pushing us toward mental shortcuts and bias.

So far, empirical evidence to settle this debate is thin. Many studies rely on self-reports or observational data that can't prove cause and effect. There is also a lack of measures to capture real-time cognitive processes. This creates a clear research gap.

We propose an experiment to close this gap. In a randomized controlled trial, we will test how AI affects decision-making. Participants will come from diverse backgrounds in age, education, and experience. Each participant will complete a series of tasks. One half will receive support via ChatGPT, a widely used AI tool. The other half will complete the same tasks without AI support for comparison.

Our method will rely on careful task design and evaluation. The decision tasks will vary in complexity through ambiguous and conflicting information. This raises mental strain. This way, we can test how AI assistance performs as the questions get harder. We measure performance through response time, task accuracy, and reasoning quality.

To assess reasoning quality, we will adapt a survey tool from Bloom and Van Reenen (2007). Originally used to measure management practices, it proved effective for scoring open-ended responses. We tailor it to fit decision-making tasks. Trained evaluators will assess responses through a double-blind process. This ensures reliable and consistent scoring across participants.

Using statistical models, we will place results into the competing frameworks. For this, we will use OLS, linear probability, and logit models with robust standard errors. The outcomes will guide interpretation. If AI improves decisions under complexity, it points to System 2 thinking. This supports Kahneman's view. In contrast, worsening decisions would offer evidence for Simon. It would suggest AI creates overload and reinforces bias.

Our results will help shape future legislation. If AI improves decisions, it could justify broader use in high-stakes areas like healthcare, courts, or public administration. But if AI introduces bias, regulation may be needed. This could involve transparency standards, oversight systems, and user training. Such insights promote responsible use and informed policy.

2 Grand Proposal

AI as a disruptive technology is rapidly reshaping our everyday lives. In 2024, 72% of companies integrated AI into at least one business function. This is a sharp rise from 55% the previous year (Statista, nd). A survey conducted in the summer of 2024 found that 32% of respondents had used generative AI in the past week. 26% of them for personal tasks and 24% for work (Levine, nd). Even classrooms are changing: according to the Pew Research Center, 25% of U.S. teens now use chatbots to support their schoolwork (Gottfried et al., 2025).

This level of integration mirrors the adoption of earlier disruptive technologies, such as the internet. But AI differs in a fundamental way. It adapts to individual users, evolves through interaction, and delivers context-sensitive output. As a result, people don't just use AI — they increasingly trust and depend on it.

Our interactions with AI are not simply input-output exchanges because the output informs our next input. And that's something new. AI is becoming part of how we learn, communicate, and form opinions. In short, it is becoming part of our socialization. It influences how we think, behave, and ultimately, how we make decisions.

Human decision-making often takes place under time pressure and with limited information. This can lead to biased or suboptimal choices, and that's where AI comes in. It can make information more abundant, relevant, and accessible. Following this line of thought, we ask: Does AI assistance lead to better or worse decision-making? Two contrasting perspectives from behavioral economics point toward different conclusions.

2.1 Theoretical Background

On the one hand, Kahneman's dual-systems theory motivates the first hypothesis. According to this framework, human thinking operates through two distinct systems. System 1 is fast, automatic, and intuitive. It handles familiar problems and routine situations. In contrast, System 2 is slow, deliberate, and analytical. It is used for tasks that require logic, calculation, or conscious reasoning (Kahneman, 2012).

Kahneman shows that people tend to rely too much on System 1. The main reason is that it is quicker and requires less mental effort. However, this reliance often occurs even when System 2 would lead to better outcomes. A key factor in this imbalance is time pressure, which leads to cognitive overload. In such situations, people fall back on mental shortcuts, known as heuristics. Richard Thaler and Cass Sunstein demonstrate in multiple studies that this leads to biased decisions. We act like humans — not like rational agents (Thaler and Sunstein, 2021).

If AI could deliver the reasoning of System 2 at the speed of System 1, it has the potential to support more rational decision-making. Chatbots can provide fast, context-sensitive output that is based on rational processes. As such, they may serve as a cognitive aid by supporting or even replacing System 1 responses in some cases. In addition, the interactive nature could help individuals recognise and counteract their heuristics.

On the other hand, the concept of bounded rationality provides a theoretical counterargument. Humans have limited cognitive capacity, which is why we rely on mental shortcuts. These heuristics serve to simplify decision making. Humans are utility satisfiers not maximisers. Instead of selecting the objectively best option, we choose one that is good enough based on our cognitive and informational constraints (Simon, 1957; Wheeler, 2024).

Even though AI can process vast amounts of data, it does not eliminate the cognitive limitations of the

human decision-maker. In other words, the bottleneck may no longer be data availability, but our ability to interpret and act on that data.

This is captured by the concept of automation bias. Humans tend to trust compelling sounding arguments, without checking them. This is true both for experts in talk shows and AI-Assistants. AI can generate a compelling argument for any point of view, depending on the input. Users may become less inclined to critically assess or verify the given explanation, instead accepting algorithmic recommendations without scrutiny (Mosier and Skitka, 1999).

Furthermore, there is the risk of information overload. AI can provide large amounts of information in a short time. While that may seem helpful, it can overwhelm users. This is especially likely when users are under time pressure or lack the expertise to filter what is relevant. This increases cognitive load and can lead to decision paralysis or reliance on superficial cues. In other words, too much information becomes counterproductive. These conditions are exactly when heuristics tend to dominate (Gross, 1964; Toffler, 1970).

In addition, AI is trained on human-generated data. This raises the risk of reproducing existing biases in human decision-making. If AI relies on records of previous decisions, its output may reflect the same flawed reasoning. The problem is that it presents biased output in a way that appears neutral or objective. These biases are embedded in the data and not always visible to users. Therefore, they are less likely to be questioned or identified.

2.2 Methodological Challenges

These contrasting theoretical predictions motivate the need for empirical testing. To assess AI's real impact, we require a research design that captures how people think and decide when interacting with AI tools.

Yet this effort comes with several well-known methodological challenges. It is inherently challenging to quantify human rationality, as it involves frequently unobservable mental processes and context-dependent decisions. What constitutes a "rational" decision is a function of the goals, constraints, and definition of the choice environment (Simon, 1957; Gigerenzer and Gaissmaier, 2010). Moreover, without random assignment, studies of real-world AI adoption are susceptible to egregious confounding: individuals who choose to use AI systematically differ from those who do not in motivation, cognitive style, or tech savvy. This self-selection vitiates causal inference.

Additionally, most studies to date utilize self-report measures, which are unable to pick up on subtle changes in the way people process information or toggle between heuristic and analytic reasoning (Kahneman, 2012). Such data are prone to recall bias and social desirability effects, further eroding the internal validity of findings. These flaws highlight the pressing need for more rigorous and controlled empirical methods.. To surmount these challenges, we propose a random assignment lab experiment and double-blind, open-ended survey.

2.3 Study Design

We will conduct a randomized controlled trial to causally estimate the effect of AI assistance on decision-making. Participants will be randomly assigned to two groups, minimizing selection bias. One group will receive treatment in the form of AI-generated suggestions during tasks. The other will function as a control group. This allows for a clear estimation of the average treatment effect of AI-intervention.

For this, we will recruit a diverse, representative sample of trial participants (n = 1,000). AI is already part of everyday routine. Over 50% of adults in developed countries use AI in daily life (McKinsey & Company, 2023). To reflect this, we include participants of different ages, education levels, and backgrounds. This strengthens the external validity of our findings.

These participants will complete a series of decision-making tasks. The tasks vary in complexity: low, medium, or high. We manipulate complexity by adding irrelevant details, conflicting cues, or ambiguous wording. This increases cognitive load. Such variations will show heuristic shortcuts and help us assess decision quality.

ChatGPT will serve as the AI assistant in our study. With over 100 million weekly users in 2024, it saw widespread adoption in decision-making (OpenAI, 2024). This makes it a fitting tool for our diverse participant sample.

We will measure the effect using three primary outcomes: response time, task accuracy, and reasoning quality. Response time will be digitally recorded in seconds. To assess task accuracy, we will use predefined answers and record it as a binary outcome (1 = correct, 0 = incorrect). Reasoning quality requires subjective evaluation. Evaluators will score open-ended responses and think-aloud protocols on a standardized 1-to-5 scale.

In particular, we assess reasoning quality using the double-blind, open-ended survey tool by Bloom and Van Reenen (2007). We tailor this tool to decision-making. Scorers will rate reasoning quality based on clarity, coherence, and logical structure. This tool also reduces bias. Interviewers will be blind to treatment, and participants to scoring criteria.

Further, we take steps to ensure consistent scoring of reasoning quality. This will include double scoring and interviewer training. A second, independent evaluator will rescore 20% of responses originally rated by someone else. This ensures inter-rater comparability. We then calculate the Intraclass Correlation Coefficient (ICC), with values above 0.75 indicating strong reliability.

To ensure robust analysis, our study will draw on interdisciplinary expertise. We will work with economists, statisticians, and behavioral scientists. Statisticians will guide data collection and test the strength of our results. Behavioral scientists will be responsible for task design. Economists will help relate findings to policy needs. Their combined input supports a study that is both rigorous and relevant.

2.4 Estimation Strategy

Our regression models include key variables to estimate the effect of AI assistance. D_i denotes the AI assistance indicator (1 = treatment, 0 = control). The coefficient on D_i, denoted β, captures the average effect of AI on the outcome, ceteris paribus. X_i is a vector of control variables. These include demographics such as age, gender, and education, and will reduce unexplained variance. Additionally, we include task complexity dummies for medium (Med) and high (High) levels. Low complexity is omitted. The interaction terms test if AI has different effects with increasing complexity.

We estimate the modeled outcomes using robust statistical methods at the task level. All models use HC3 robust standard errors. To address heteroskedasticity and within-subject correlation, we cluster standard errors at the participant level.

Balancing tests will verify whether treatment and control groups are comparable at baseline on X_i. The method depends on variable type. We use two-sided t-tests to compare group means for continuous variables. For binary or categorical variables, we apply proportion tests.

Hypothesis testing will focus on the significance of the AI treatment effect. For all models, we test the null hypothesis $H0 : \beta = 0$ versus the alternative $H1 : \beta \neq 0$. For interaction terms, we also test $H0 : \theta_1 = 0, \theta_2 = 0$.

2.5 Model Specifications

Regarding the causal effect of AI assistance on response time, we use Ordinary Least Squares (OLS), as the outcome is continuous. The regression model is:

$$ResponseTime_i = \alpha + \beta D_i + \delta_1 \text{Med}_i + \delta_2 \text{High}_i + \theta_1 (D_i \times \text{Med}_i) + \theta_2 (D_i \times \text{High}_i) + \gamma X_i + \varepsilon_i \tag{1}$$

Similarly, we estimate reasoning quality. Although measured on an ordinal 1–5 scale, we treat this outcome as continuous for simplicity. Since we are interested in average group differences, the coefficients are interpretable. Thus, we use OLS:

$$ReasoningScore_i = \alpha + \beta D_i + \delta_1 \text{Med}_i + \delta_2 \text{High}_i + \theta_1 (D_i \times \text{Med}_i) + \theta_2 (D_i \times \text{High}_i) + \gamma X_i + \varepsilon_i \tag{2}$$

We take a different statistical approach when analyzing accuracy. This outcome is binary: correct = 1, incorrect = 0. Therefore, we use an OLS Linear Probability Model and a logit model for robustness. The LPM offers simple and interpretable coefficients, while the logit accounts for the binary nature of the outcome. Consequently, we will report average marginal effects (AME). These are expressed as percentage point changes in the probability of correct responses.

$$Correct_i = \alpha + \beta D_i + \delta_1 \text{Med}_i + \delta_2 \text{High}_i + \theta_1 (D_i \times \text{Med}_i) + \theta_2 (D_i \times \text{High}_i) + \gamma X_i + \varepsilon_i \tag{3}$$

$$\text{Logit}(\Pr(Correct_i = 1)) = \alpha + \beta D_i + \delta_1 \text{Med}_i + \delta_2 \text{High}_i + \theta_1 (D_i \times \text{Med}_i) + \theta_2 (D_i \times \text{High}_i) + \gamma X_i \tag{4}$$

2.6 Interpretation and Implications

We will interpret results within the frameworks of Kahneman and Simon. $\beta > 0$ for reasoning quality and response time would indicate AI promoting slow, analytical processing. The same applies to a positive AME for correct responses. If we see interaction effects where $\theta_2 > \theta_1 > 0$, AI would be more beneficial under cognitive load. These patterns would support Kahneman's view. Conversely, $\beta < 0$, a negative AME, and $\theta_2 < \theta_1 < 0$ hint at heuristic shortcuts as task difficulty rises. This would align with Simon's theory.

These findings will help clarify AI's role in human decision-making and lay the groundwork for evidence-based policy. If AI improves thinking, then the government must use more AI in the public sector, hospitals, and courts to get work done faster and with fewer errors. If AI encourages greater bias, then legislators must pass legislation to make AI systems more transparent. They can also set up review panels to screen AI decisions for fairness and increase training for those who utilize AI programs. Our research will underscore whether AI facilitates or hinders decision-making. Our report will guide future legislation and lead responsible application of AI in society.

3 Roles and Responsibilities

Our project is a collaborative international effort between York University and LMU Munich. The study will be carried out on both university campuses. The study is coordinated by our tutor and project lead, **Dr. Evelina Griniute**, who ensures smooth collaboration between two core teams:

- **Core Team 1 (York University): Taha Ranjbar** (Political Science & Public Policy), **Yuehan An** (Economics & Data Science), **Senugi Jayawardana** (Law & Governance)
- **Core Team 2 (LMU Munich): Noah Mund** (Philosophy & Behavioral Economics), **Livia Hess** (Econometrics & Statistics)

Each team is responsible for running experiments on their respective campuses. Tasks include recruiting interviewers for double-blind interviews, conducting sessions, and collecting data. Dr. Griniute selected each team member to ensure a broad interdisciplinary skill set. Accordingly, each member takes on a specific task that reflects their field of study and applied strengths:

- **Evelina Griniute (Project Head – Social Psychology & Policy):** Her task is to coordinate between York and LMU and oversee all project milestones. In particular, she ensures that all deliverables meet the highest academic and ethical standards.
- **Noah Mund (Theory Lead – Philosophy & Behavioural Economics):** He is tasked with developing the theoretical framework in consultation with advisors. His focus lies on integrating insights from dual-systems theory and bounded rationality. He authors the conceptual and theory-driven sections of the papers.
- **Livia Hess (Methodology Lead – Econometrics & Statistics) & Yuehan An (Quantitative Analyst – Economics & Data Science):** Together, they design the experimental protocol and develop the random assignment structure. Furthermore, they conduct the empirical analysis, including regression models and robustness checks. They also co-author the methodology section of the paper.
- **Senugi Jayawardana (Policy Analyst – Law & Governance) & Taha Ranjbar (Strategic Policy – Political Science & Public Policy):** They have experience in both public administration and legislative analysis. Their role is to translate the project's empirical insights into actionable policy recommendations. Therefore, they lead the writing of the policy impact section. In addition, they help position the research within real-world governance challenges.

We also secured the external guidance of Professors Richard Thaler and Cass Sunstein. They are internationally recognized for their work in behavioural economics and experimental design. We are grateful that they agreed to support the experimental setup and help avoid common design fallacies.

We aim to fully use the interdisciplinary advantage of our team. We combine perspectives from economics, philosophy, data science, law, psychology, and political science. At the same time, we rely on clearly defined responsibilities and complementary expertise. This interdisciplinary structure positions our team to execute the project with both scientific rigor and real-world relevance.

4 Knowledge Mobilization Plan

To disseminate the research results to both academic and non-academic stakeholders, we will take a multi-channel knowledge mobilization strategy.

First, we will report the findings in leading academic conferences such as the Annual Meeting of the American Economic Association (AEA) and the Behavioural Economics Annual Meeting (BEAM), which are rigorous channels for feedback from economists and behavioral scientists. Since the project is interdisciplinary, conferences such as the ACM Conference on Fairness, Accountability, and Transparency (FAccT) or NeurIPS (for its social implications track) may also be suitable venues to communicate with researchers in AI and policy.

We will target publication in peer-reviewed outlets such as the *Journal of Economic Behavior and Organization, Nature Human Behaviour*, or *AI & Society*, as they are well-positioned to reach economic theorists and researchers of the social consequences of AI.

In addition to academic publication, we will also develop a brief policy brief outlining the findings and implications for governance decision-making that can be circulated through think tanks or distributed to policymakers through channels such as the Brookings Institution or OECD AI Observatory.

If the results are susceptible to broader public interest, we are happy to organize a mini symposium or panel discussion—perhaps in collaboration with a university research center or interdisciplinary AI ethics lab—to bring together scholars, practitioners, and policymakers. This would also be a chance to discuss follow-on questions and translational potential.

Finally, we are happy to utilize online forums such as *The Conversation* or *Medium* to translate main findings into plain-language narratives for a general audience.

5 Expected Outcomes

Scholarly Results:

Our project will test if AI causes people to think more carefully or forces them to rely on quick hunches. We will do this by performing an experiment in which participants work with AI or without AI, and we will compare how they make decisions. This method blends random assignment, interviews, and open-ended questions.

Our findings will be adding to research in behavioral economics, notably ideas by Daniel Kahneman (who writes of rational, "System 2" thinking) and Herbert Simon (who speaks to why humans use cognitive shortcuts). If we find that AI leads people to think more analytically, it validates Kahneman's theory. If we find that AI leads to additional shortcuts, it validates Simon's theory. By releasing our findings and design, we are also giving future researchers a method for gauging the ways in which AI changes people's thinking.

Societal Implications:

Our results can guide government and business applications of AI. If AI improves prudent thinking, policymakers can champion the application of AI in state houses, clinics, or courthouses to speed service and reduce mistakes.

If AI stimulates shortcuts or bias, we might need new regulations that require businesses to reveal how their AI works. We might also need training courses so people can recognize biased or incorrect AI choices. Either way, our research will allow society to utilize AI more responsibly, so its benefits do not come with larger risks or underlying problems.

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